

UpSkiller

AI-Powered Personalized Learning Roadmap

Mobile Application

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Academic Info

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About UpSkiller

UpSkiller is an AI-powered mobile application that generates personalized learning roadmaps based on a user's existing skills and desired learning goals. Unlike traditional one-size-fits-all roadmaps, UpSkiller creates customized learning paths by analyzing what you already know, skipping topics you have already mastered, and focusing exclusively on the knowledge gaps that matter most to you.

The core idea is simple: every learner is different. Two developers aiming to learn Docker may already know vastly different things. UpSkiller accounts for this by using AI to build a unique roadmap for each individual, making learning more efficient and less repetitive.

Traditional Roadmaps

One roadmap for everyone
Teaches topics you already know
Wastes your time

UpSkiller

Different roadmap for every user
Skips what you already know
Saves your time

Tech Stack

Layer	Technology
Backend	Django, Django REST Framework (DRF)
AI Layer	LangGraph + Gemini 2.5 Flash
Frontend	Native Android (Java)

Database	SQLite (Development) / PostgreSQL (Production)
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Repository Structure

```
upskiller/  
├── backend/           # Django REST API + LangGraph AI agent  
├── android_app/       # Native Android application (Java)  
└── documentation.docs # Project documentation & SRS
```

Complete Working Flow

Step 1 — User Registration

A new user signs up by providing their name, email, and password. An optional profile picture can also be uploaded.

Immediately after registration, the app prompts the user to build their skill profile by asking: "What skills do you already have?"

Example initial skill profile:

Skill	Level
Python	Advanced
FastAPI	Intermediate
Docker	Beginner

Users can add, remove, or update skill levels at any time from their profile.

Step 2 — Generate a New Roadmap

When a user wants to learn something new, they simply enter:

- Skill to Learn — e.g., MCP Servers
- Target Level — Beginner, Intermediate, or Advanced

Step 3 — AI Personalization

The system sends the user's full skill profile along with the target skill and desired level to the AI engine (LangGraph + Gemini 2.5 Flash). The AI analyzes:

- What the user already knows
- Which prerequisite topics can be safely skipped
- Which missing topics are required to reach the target level

Step 4 — Structured AI Response

The AI does not return a chat message. It returns structured JSON data only, which the backend stores in the database and the frontend renders as a visual roadmap.

Example AI JSON response:

```
{
  "roadmap_title": "MCP Servers",
  "target_level": "Advanced",
  "phases": [
    {
      "name": "Foundation",
      "topics": [
        {"title": "MCP Fundamentals"},
        {"title": "Tool Calling"}
      ]
    },
    {
      "name": "Intermediate",
      "topics": [
        {"title": "Transport Protocols"},
        {"title": "Authentication"}
      ]
    },
    {
      "name": "Advanced",
      "topics": [
        {"title": "Production Deployment"}
      ]
    }
  ]
}
```

Step 5 — Visual Roadmap on Android

The Android app renders the JSON as a structured visual roadmap, not chat bubbles. Users see phases and topics organized in a clear, trackable layout:

```
FOUNDATION
☐ MCP Fundamentals
☐ Tool Calling

INTERMEDIATE
☐ Transport Protocols
☐ Authentication

ADVANCED
☐ Production Deployment
```

Step 6 — Topic Completion

Users mark individual topics as completed. Each topic has an `is_completed` flag that updates in real time.

```
FOUNDATION
☒ MCP Fundamentals (completed)
```

✓

Tool Calling

(completed)

INTERMEDIATE

☐

Transport Protocols

☐

Authentication

Step 7 — Automatic Roadmap Completion

When every topic in every phase is marked complete, the roadmap status automatically changes to COMPLETED. There is no manual "Complete Roadmap" button — the system handles this automatically.

Step 8 — Automatic Skill Addition

Once a roadmap is completed, the system automatically adds the newly learned skill to the user's profile at the target level.

Example: Completing the MCP Servers → Advanced roadmap automatically adds MCP Servers → Advanced to the user's skill profile. This enriched profile makes all future AI-generated roadmaps even more personalized.

Step 9 — Multiple Active Roadmaps

Users can run multiple roadmaps simultaneously. For example:

- Django — In Progress
- Docker — In Progress
- MCP Servers — In Progress

Step 10 — Roadmap History

The app maintains a full history of all roadmaps. Nothing is ever deleted.

In Progress	Completed
Django Docker	Git Python MCP Servers

Functional Requirements

ID	Feature	Description
FR-1	User Registration	Register with name, email, password, and optional profile picture.
FR-2	User Login	Authenticate with email and password.

FR-3	Skill Setup	Prompt user to add existing skills after registration.
FR-4	Skill Management	Add, update, delete, and view skills at any time.
FR-5	Roadmap Generation	Submit target skill and level to generate a roadmap.
FR-6	AI Personalization	AI skips known topics and generates only missing content.
FR-7	Structured Response	AI returns structured JSON, not conversational text.
FR-8	Roadmap Storage	All generated roadmaps are persisted in the database.
FR-9	Multiple Roadmaps	Users can have multiple roadmaps active simultaneously.
FR-10	Topic Completion	Users mark individual topics as complete.
FR-11	Auto-Completion	Roadmap status updates automatically when all topics are done.
FR-12	Skill Auto-Add	Completed roadmap target skill is auto-added to user profile.
FR-13	Roadmap History	Users can view active and completed roadmaps.
FR-14	Profile Management	Users can update name, profile picture, and skills.

Database Schema

Table	Key Fields	Purpose
User	id, name, email, password, profile_picture	Stores registered user accounts.
Skill	id, user(FK), skill_name, level	Stores user's skill profile entries.
Roadmap	id, user(FK), title, target_level, status, created_at	Stores generated roadmaps.
Phase	id, roadmap(FK), phase_name, order	Groups topics into learning phases.
Topic	id, phase(FK), title, is_completed	Individual learning tasks within a phase.

System Architecture



Non-Functional Requirements

Category	Requirement
Performance	Normal API operations complete within 2 seconds.
Reliability	User data and roadmaps persist across app restarts.
Usability	Simple interface with clear roadmap visualization and easy progress tracking.
Maintainability	Modular architecture for straightforward future enhancements.
Scalability	Backend supports future migration from SQLite to PostgreSQL.

Constraints

- An active internet connection is required for AI roadmap generation.
- Roadmap generation depends on Gemini API availability.
- Personalization quality depends on the accuracy of user-provided skills.

Future Enhancements

- Resource recommendations (articles, documentation links)
- Video link integration per topic
- Progress percentage tracking
- Search functionality across roadmaps and topics
- Push notifications for learning reminders
- AI mentor chat for contextual help
- Skill analytics and learning statistics
- Streak system to encourage daily learning

- Dark mode support